

Lab 7 – Investigating DNS

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INTRODUCTION

In this lab, I will be using my command prompt and Wireshark to complete 5 goals regard DHC and DNS. First, I will use my command prompt to find my DHCP/DNS specifications and using various command to navigate those packets. I will then analyze those packets with Wireshark.

BREAKPOINT 1

1. What is the role of DHCP and DNS in your access to the Internet?

DHCP gives my device an IP address and certain network settings and DNS translate domain names into IP addresses so you can reach websites.

2. Is DHCP enabled? Is it autoconfigured? How long is the DHCP lease?

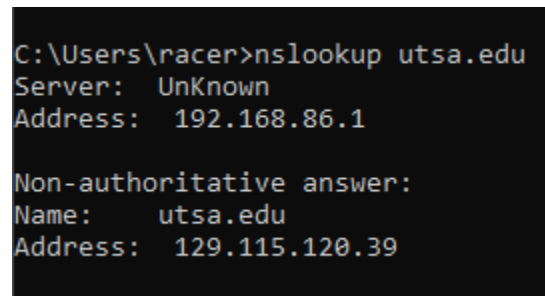
Yes, and yes. The DHC lease is about 3 days.

3. What is the IP address of the DHCP and DNS servers?

DHCP/DNS IP: 192.168.86.1

BREAKPOINT 2

Below I used the nslookup command for utsa.edu. It comes back with two sets of info (server/address and name/address) that gives details about this domain.



```
C:\Users\racer>nslookup utsa.edu
Server:  UnKnown
Address:  192.168.86.1

Non-authoritative answer:
Name:     utsa.edu
Address:  129.115.120.39
```

Figure 1: nslookup results for utsa.edu

BREAKPOINT 3

Here I used the command nslookup -type=ns for utsa.edu. As shown below, the authoritative server shows the server/address while the non-authoritative server shows the name of the server.

```

C:\Users\racer>nslookup -type=ns utsa.edu
Server: UnKnown
Address: 192.168.86.1

Non-authoritative answer:
utsa.edu      nameserver = ns2.utsa.edu
utsa.edu      nameserver = ns1.utsa.edu
utsa.edu      nameserver = ns3.utsa.edu

```

Figure 2: results for nslookup and auth/non-auth responses

BREAKPOINT 4

I performed a reverse DNS search using my command prompt by using the set q=ptr command and entering the IP address for utsa.edu. AS shown below the command successfully found the server and displayed the name of it.

```

C:\Users\racer>nslookup
Default Server: UnKnown
Address: 192.168.86.1

> set q=ptr
> 129.115.120.39
Server: UnKnown
Address: 192.168.86.1

Non-authoritative answer:
39.120.115.129.in-addr.arpa    name = utsa.edu

```

Figure 3: results of reverse dns search

BREAKPOINT 5

1. The packet number for the initial canvas query is 2996 and the packet was transported via UDP.
2. The packet number for the canvas response is 2997 and the packet was transported via UDP.
3. The destination port for the request is 53 and the source port for the response is 53.
4. The DNS query was sent to my IP address (192.168.86.1).
5. The request has only 1 question and 0 answers.
6. The query reply has 1 question and 2 answers. The two answers it received provides the URL, type A, class IN, and two different IP addresses (104.20.39.17 and 172.66.158.225). The first answer also contains the name (utsa-canvas.easysoft.com), type (A), class (IN (0x0001)), time to live (5 min), and data length (4).

CONCLUSION

Replace the text in this section with a reflective summary of your process in this lab, including answers to the following, in complete sentences:

- What challenges did you encounter and/or what most interested you about this lab?
I had some trouble figuring out what I was looking for regarding the query request and response packets.

- How did you address your challenges and/or what additional research did you do?
After some clicking around, I quickly figured out how they're connected and what the instructions were asking me to find.
- How might you use the skills and concepts of this lab going forward?
It grants me an even deeper understanding about how query requests/responses work if there's no cache within a system to navigate to a server as well as how they communicate with the desired server.

REFERENCES

R. Mitra, "*Lab 7: Investigating DNS*," The University of Texas at San Antonio (2024). Last accessed: 4/28/26

GENERATIVE AI SEARCHES

I didn't use AI

COLLABORATION

I worked alone.